B.Tech. (Computer Science and Engineering)

Semester – VII
Subject: Data Mining and Data Warehousing (BCO029B)
Assignment #5

SECTION – A

(5 × 2 = 10 Marks)

Q1. Define partitioned clustering in data mining and give one example algorithm.

Answer:
Partitioned clustering is a clustering technique in data mining that divides a dataset into a fixed number of non-overlapping clusters. Each data object belongs to exactly one cluster, and the goal is to optimize an objective function such as minimizing intra-cluster distance.

Example Algorithm:

- **K-Means Clustering**

Q2. What is a dendrogram in hierarchical clustering?

Answer:
A dendrogram is a tree-like diagram used in hierarchical clustering to represent the arrangement of clusters formed during the clustering process. It shows how individual data points are merged into clusters at different similarity levels.

Q3. Define Web content mining.

Answer:
Web content mining is the process of extracting useful information from the contents of web pages. This content may include text, images, audio, video, and structured data such as tables and lists.

Q4. What is Web structure mining? Mention one typical application.

Answer:
Web structure mining analyzes the hyperlink structure of the web to understand relationships between web pages.

Typical Application:

- **Page ranking**, such as Google's PageRank algorithm.

Q5. Write any two challenges in multimedia data mining.

Answer:

1. Handling large volume and high dimensional multimedia data
2. Difficulty in extracting meaningful features from images, audio, and videos

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SECTION – B

(3 × 7 = 21 Marks)

Q1. Explain density-based clustering. Describe the concepts of core, border, and noise points and outline the working of DBSCAN.

Answer:
Density-based clustering groups data points based on regions of high density separated by regions of low density. It is effective in identifying clusters of arbitrary shapes and handling noise.

Key Concepts:

- **Core Point:** A point with at least MinPts neighbors within distance ε.
- **Border Point:** A point that is within ε distance of a core point but has fewer than MinPts neighbors.
- **Noise Point:** A point that is neither a core nor a border point.

DBSCAN Working:

1. Select an unvisited point.
2. Check if it is a core point.
3. If yes, form a cluster by expanding neighbors.
4. Repeat until all points are processed.
5. Noise points remain unclustered.

Q2. Explain the basic process of text mining. Discuss text preprocessing, document representation, and at least two text mining tasks such as classification and sentiment analysis.

Answer:
Text mining extracts meaningful information from unstructured textual data.

Text Preprocessing:

- Tokenization
- Stop-word removal
- Stemming / Lemmatization

Document Representation:

- Bag of Words (BoW)
- TF-IDF (Term Frequency–Inverse Document Frequency)

Text Mining Tasks:

1. **Text Classification:** Assigning documents to predefined categories (e.g., spam detection).
2. **Sentiment Analysis:** Identifying emotions or opinions (positive, negative, neutral).

Q3. What are grid-based clustering methods? Explain the general approach and discuss any one algorithm such as STING or CLIQUE.

Answer:
Grid-based clustering methods divide the data space into a finite number of grid cells and perform clustering on these cells instead of individual points.

General Approach:

1. Divide data space into grids.
2. Compute density for each grid.
3. Merge dense grids to form clusters.

STING Algorithm:

- Uses hierarchical grid structure.
- Stores statistical information in each grid.
- Efficient for large datasets.

SECTION – C

(3 × 11 = 33 Marks)

Q1. Discuss Web mining in detail. Clearly define Web content mining, Web structure mining, and Web usage mining, explaining sources of data, typical methods, and practical applications for each category.

Answer:
Web mining is the application of data mining techniques to discover patterns from web data.

1. Web Content Mining

- **Data Sources:** Web pages, blogs, multimedia content
- **Methods:** NLP, information extraction
- **Applications:** Search engines, recommendation systems

2. Web Structure Mining

- **Data Sources:** Hyperlinks between web pages
- **Methods:** Graph theory, link analysis
- **Applications:** Page ranking, community detection

3. Web Usage Mining

- **Data Sources:** Server logs, cookies, clickstream data
- **Methods:** Pattern discovery, association rules
- **Applications:** Personalized recommendations, user behavior analysis

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Q2. Write detailed notes on spatial and multimedia data mining. Describe data characteristics, representative techniques, applications, and major research challenges.

Answer:

Spatial Data Mining

- **Characteristics:** Location-based, spatial relationships
- **Techniques:** Spatial clustering, spatial association rules
- **Applications:** GIS, weather forecasting, urban planning
- **Challenges:** Spatial autocorrelation, large data volume

Multimedia Data Mining

- **Characteristics:** High dimensional, heterogeneous
- **Techniques:** Feature extraction, deep learning
- **Applications:** Medical imaging, video surveillance
- **Challenges:** Semantic gap, storage and processing complexity

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Q3. Elaborate the complete pipeline of text mining with a suitable case study (e.g., sentiment analysis of social media data).

Answer:

Text Mining Pipeline:

1. **Data Collection:** Tweets, reviews, posts
2. **Preprocessing:** Cleaning, tokenization, stemming
3. **Feature Extraction:** TF-IDF, word embeddings
4. **Model Building:** Naïve Bayes, SVM, LSTM
5. **Evaluation:** Accuracy, precision, recall

6. **Extensions:** Emotion detection, topic modeling

Case Study – Sentiment Analysis:
Analyzing customer tweets to classify opinions as positive, negative, or neutral helps companies improve products and services.
